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KGiSL Campus, Saravanampatti, Coimbatore – 641 035

JAVA PROJECT REPORT ON

Library Book Borrowing, Return and Fine Calculation System

Submitted by

MANGALESH M

25BIT031

Under the Guidance of

DHILEEBAN A

Technical Trainer

Technical Department, KG College of Arts and Science

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Semester: **I**

DECLARATION

I hereby declare that the project entitled “Library Book Borrowing, Return and Fine Calculation System**”** submitted in partial fulfillment of the requirements for the course [Java Programming **]** is a **record of my original work** carried out under the guidance of DHILEEBAN A **,** Technical Trainer, Technical Department, KG College of Arts and Science.

I further declare that this project has not been submitted previously, either in part or in full, for the award of any degree, diploma, or other similar title, at this or any other institution.

I have acknowledged all sources of information used in this project wherever applicable.

April 2026

Coimbatore

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MANGALESH M

CERTIFICATE

This is to certify that the project entitled **"Library Book Borrowing, Return and Fine Calculation System"** is a **bona fide work** carried out by **MANGALESH M** , B.Sc IT I-'A' Information Technology, **KG College of Arts & Science**, in partial fulfillment of the requirements for the course **[Java Programming]**.

This project has been completed under my supervision and guidance during the academic period **April 2026**.

Place: Coimbatore

Date:09.04.2026

Submitted for the Viva-Voce Examination held on _____

Guide:
(Technical Trainer, KGCAS)

_____

Head of the Department
(Technical Head, KGCAS)

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ABSTRACT

The **Library Book Borrowing, Return and Fine Calculation System** is a console-based application developed using the Java programming language to automate the process of issuing and returning books in a library environment. In many educational institutions and small libraries, book issuing and returning activities are handled manually, which increases the chances of errors and delays in maintaining records.

This project provides a simple and efficient solution where users can borrow books by entering their name, book title, and borrowing duration. When the book is returned, the system checks whether the return is delayed and calculates the fine automatically based on the number of extra days.

The system is implemented using core Java concepts such as classes, objects, methods, loops, switch statements, and conditional statements. The application provides a menu-driven interface that allows users to perform operations easily.

The main objective of this system is to reduce manual work, improve accuracy, and demonstrate the practical implementation of object-oriented programming concepts in Java.

1. INTRODUCTION

Libraries are essential parts of educational institutions because they provide access to academic books and reference materials required by students and faculty members. Managing library operations manually becomes difficult when the number of books and users increases.

Traditional library systems maintain records using notebooks or registers. This manual method consumes more time and may lead to errors such as incorrect fine calculation, missing records, and difficulty tracking borrowed books.

To overcome these issues, a computerized system is necessary. The **Library Book Borrowing, Return and Fine Calculation System** helps automate library operations by allowing users to borrow books, return books, and calculate fines automatically if the books are returned after the due date.

This system improves efficiency, accuracy, and speed of library management. It also helps students understand how Java programming can be used to develop real-world applications.

1.1 Objective

The main objectives of this project are:

• To design a simple library borrowing system using Java
• To automate the book borrowing process
• To automate the book return process
• To calculate late return fine automatically
• To reduce manual calculation errors
• To provide a user-friendly menu-driven system
• To demonstrate object-oriented programming concepts
• To improve efficiency in managing library operation

1.2 Overview

The system is developed as a menu-driven console application using Java. It allows the user to interact with the system through different options displayed on the screen.

The user can select operations such as:

1. Borrow Book
2. Return Book
3. Exit System

During borrowing, the system collects:

- Student name
- Book name
- Borrowing duration

During returning, the system:

- Accepts return duration
- Compares borrowing duration
- Calculates fine if delayed

The system ensures that library transactions are handled quickly and accurately.

1.3 Features

Main features of the system include:

Borrow Book Feature

Allows users to enter student details and book details before issuing the book.

Return Book Feature

Allows users to return borrowed books by entering return duration.

Automatic Fine Calculation Feature

Calculates fine automatically if the return exceeds allowed borrowing days.

Example:

If borrowing days = 5
Return days = 7

Fine = 2 × 5 = Rs.10

Menu-Driven Interface

Displays simple options for easy interaction.

Error Reduction

Minimizes manual mistakes during calculations.

2. PROGRAM ANALYSIS

Manual library management systems create several difficulties such as:

- Maintaining borrowing records manually
- Difficulty tracking return dates
- Errors in calculating fines
- Increased paperwork
- Time-consuming operations

Because of these problems, an automated system is necessary to improve efficiency and accuracy.

The proposed system solves these problems by providing an automated borrowing and returning process.

2.1 Problem Definition

Manual library management systems create several difficulties such as:

- Maintaining borrowing records manually
- Difficulty tracking return dates
- Errors in calculating fines
- Increased paperwork
- Time-consuming operations

Because of these problems, an automated system is necessary to improve efficiency and accuracy.

The proposed system solves these problems by providing an automated borrowing and returning process

2.2 Project Overview

The system is developed using Java programming language and performs basic library operations through a console interface.

The system performs the following tasks:

Accepts student name
Accepts book name
Stores borrowing duration
Accepts returning duration
Calculates fine automatically

This project demonstrates how object-oriented programming can be used to develop simple real-world applications

2.3 Scopes and Limitations

Scope

The system can be used in:

• School libraries
• College libraries
• Tuition centers
• Training institutes

Functions supported:

Borrow book
Return book
Calculate fine

Limitations

Current system limitations include:

Supports only one book at a time
No database connectivity
Console-based interface only
No record storage facility

Future Enhancements

System can be improved by adding:

Database storage
Multiple book handling
Graphical user interface
Student ID tracking
Automatic due date calculation

3. SYSTEM DESIGN

System design describes how the system is structured into modules.

3.1 MODULES (Detailed Explanation)

Module 1: Borrow Book Module

This module collects information from users.

Inputs:

Student name
Book name
Borrow days

Example:

Enter Student Name
Enter Book Name
Enter Borrow Days

Output:

Book Borrowed Successfully

Module 2: Return Book Module

This module collects return duration from user.

Input:

Return days

Example:

Enter Number of Days Taken to Return Book

Output:

Book Returned Successfully

Module 3: Fine Calculation Module

This module compares borrow days and return days.

If delayed:

Fine calculated using formula:

Fine = (Return Days − Borrow Days) × 5

Otherwise:

Fine = 0

Module 4: Exit Module

Terminates program execution safely.

4. METHODOLOGY

4.1. Problem Analysis

Manual library operations require more time and effort. Therefore, automation improves system efficiency and reduces calculation errors.

4.2. System Design

System structure includes:

LibrarySystem class
LibraryMain class

Methods used:

borrowBook()
returnBook()

4.3. Program Development

The program is developed using Core Java concepts such as:

Classes

Objects

Scanner class

Methods

Switch statement

Conditional statements

Example:

```
if(returnDays > borrowDays)
```

Used for checking delay condition.

Example:

```
switch(choice)
```

Used for menu selection..

4.4. Testing and Validation

Test Case	Input	Expected Output
Borrow book	Enter student name	Book borrowed successfully
Return book early	Return ≤ borrow days	No fine
Return book late	Return > borrow days	Fine calculated
Exit option	Press 3	Program terminated

4.5. Result and Documentation

The system was tested successfully with different input values.

Results show:

Borrow operation works correctly

Return operation works correctly

Fine calculation works correctly

The documentation explains system structure and implementation clearly

4.6. Algorithm

Step 1: Start program
Step 2: Display menu options
Step 3: Accept user choice
Step 4: If choice = Borrow Book
→ Enter student name
→ Enter book name
→ Enter borrow days

Step 5: If choice = Return Book
→ Enter return days

Step 6: Compare return days with borrow days

Step 7: If delayed
→ Calculate fine

Step 8: Display fine amount

Step 9: Repeat process until exit selected

Step 10: Stop program

4.7. Flowchart

Students should draw a flowchart showing program flow.

Example flow:

Start

↓

Display Menu

↓

Enter Choice

↓

Borrow Book

↓

Return Book

↓

Check Delay

↓

Calculate Fine

↓

Display Fine

↓

Exit

5. SOURCE CODE

```
import java.util.Scanner;

class LibrarySystem {

    String bookName;
    String studentName;
    int borrowDays;
    int fine = 0;

    void borrowBook() {
        Scanner sc = new Scanner(System.in);

        System.out.println("Enter Student Name:");
        studentName = sc.nextLine();

        System.out.println("Enter Book Name:");
        bookName = sc.nextLine();
```

```
        System.out.println("Enter Number of Days to Borrow:");
        borrowDays = sc.nextInt();

        System.out.println("Book Borrowed Successfully!");
    }

    void returnBook() {
        Scanner sc = new Scanner(System.in);

        int returnDays;

        System.out.println("Enter Number of Days Taken to Return Book:");
        returnDays = sc.nextInt();

        if (returnDays > borrowDays) {
            fine = (returnDays - borrowDays) * 5;
        } else {
            fine = 0;
        }

        System.out.println("Book Returned Successfully!");
        System.out.println("Fine Amount = Rs. " + fine);
    }
}

public class LibraryMain {

    public static void main(String[] args) {
```

```
Scanner sc = new Scanner(System.in);
LibrarySystem lib = new LibrarySystem();

int choice;

do {
    System.out.println("\n--- Library Menu ---");
    System.out.println("1. Borrow Book");
    System.out.println("2. Return Book");
    System.out.println("3. Exit");

    System.out.println("Enter your choice:");
    choice = sc.nextInt();

    switch (choice) {

        case 1:
            lib.borrowBook();
            break;

        case 2:
            lib.returnBook();
            break;

        case 3:
            System.out.println("Thank you for using Library System");
            break;
```

```
            default:
                System.out.println("Invalid Choice");
        }

    } while (choice != 3);

  }
}
```

6. INPUT AND OUTPUT SCREENS



7. DISCUSSION & CONCLUSION

7.1 Discussion

The system successfully performs borrowing and returning operations using a menu-driven interface. It calculates fine automatically based on delay duration.

The project demonstrates the use of object-oriented programming concepts such as classes and methods effectively.

Testing results show that the system works correctly under different conditions.

7.2 Conclusion

The **Library Book Borrowing, Return and Fine Calculation System** is successfully developed using Java programming language.

The system:

Reduces manual workload
Improves calculation accuracy
Provides faster operations
Demonstrates practical Java concepts

Future improvements can include database connectivity and graphical interface support

8. REFERENCES

The following resources were used during the development of this project:

Books

Books

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